

**MTD215**  
**Application of C++ in Multimedia**

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**Tutor-Marked Assignment**

**July 2023 Presentation**

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***TUTOR-MARKED ASSIGNMENT (TMA)***

This assignment is worth 15% of the final mark for MTD215 – Application of C++ in Multimedia.

The cut-off date for this assignment is **9 October 2023, 23:55 hrs.**

**Note to Students:**

You are to include the following particulars in your submission: Course Code, Title of the TMA, SUSS PI No., Your Name, and Submission Date.

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**Question 1**

- (a) Based on what you have learnt in C++, explain how Object-Oriented Programming (OOP) principles are implemented in C++. Do remember the four major principles of OOP and use C++ examples to illustrate those four points. (12 marks)
- (b) Based on what you have learnt in the course and explored online, appraise and discuss SDL for graphic surfaces and multimedia applications by comparing with other APIs or libraries (Choose **THREE (3)** for comparison). List out both the advantages and disadvantages of SDL.

| API/Libraries | Pros | Cons |
|---------------|------|------|
| SDL           | ...  |      |
| ...           |      |      |

(13 marks)

**Question 2**

Apply C++ to program a singing competition scoring system. Remember to use comments to explain the syntax and structure of C++ language.

Requirements of the scoring system:

- There are 10 judges to score the contestants.
- Each judge gives a score between 0 and 10 points (inclusive).
- The final score is the average of the remaining 8 scores after removing the highest and the lowest score.

Write down codes with comments to implement the following functions:

- (a) Receive the score of each judge. (10 marks)
- (b) Identify and remove the highest and lowest scores. (10 marks)
- (c) Calculate and display the final average score. (5 marks)

### Question 3

Implement a C++ program in an objected-oriented programming style and use comments to explain the syntax and structure of C++.

- (a) Declare the `Circle` class as follows:
  - (i) Two private variables: Radius and Pi. (2 marks)
  - (ii) Constructor with input of radius. (2 marks)
  - (iii) One function to get perimeter. (2 marks)
  - (iv) Another function to get diameter. (2 marks)
  - (v) Another function to get area. (2 marks)
- (b) Write a `main()` function to demonstrate how the constructors in part (a) are being used.
  - (i) To get user inputs of radius. (2 marks)
  - (ii) Create an object based on user inputs. (2 marks)
  - (iii) Display the perimeter, diameter and area of the circle. (6 marks)
- (c) Using SDL or SFML to visualize the circle. (5 marks)

**Question 4**

- (a) Apply C++ and SDL to implement a framework of SNAKE game:
- (i) Create a window with resolution of 800\*600.
  - (ii) Display a red block with resolution of 50\*50.
  - (iii) Use keyboard ASWD or arrow keys to move the red block.

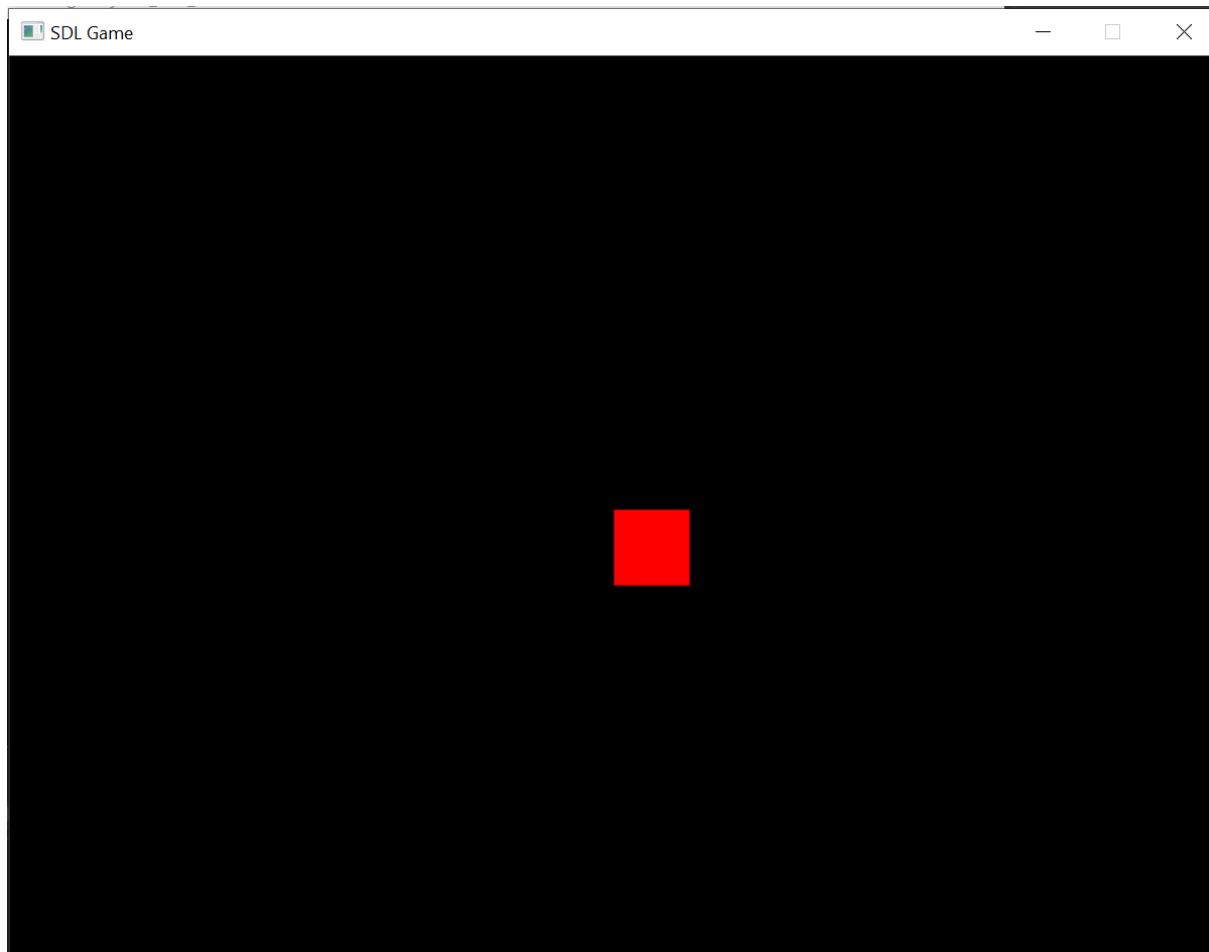


Figure Q4: Output of SDL program

(20 marks)

- (b) Appraise the use of SDL for this kind of classic gaming. Pick **ONE (1)** game developed using SDL and discuss how SDL was used in the game's development and what specific features of SDL were particularly beneficial for this game.

(5 marks)

---- END OF ASSIGNMENT ----